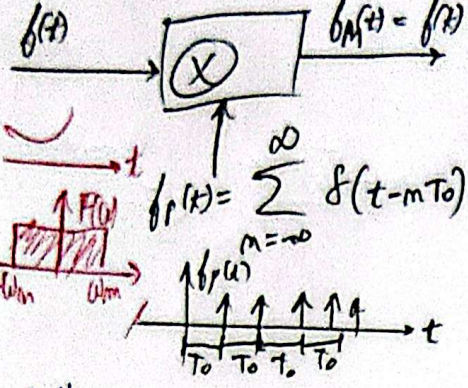
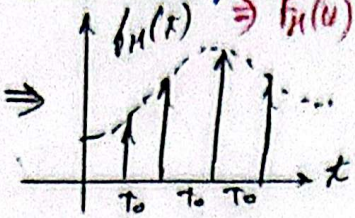
* **Modulación PAM Ideal** = Se considera lo más ideal posible, la señal de muestreo será un Tren de Pulsos de Dirac.



$f_M(t) = f(t)\cdot f_p(t) \Rightarrow$



$\Rightarrow F_M(\omega)$

$$\Rightarrow \boxed{\begin{array}{l} f(t)\cdot g(t) \xrightarrow{\mathcal{F}} F(\omega)\,G(\omega) \\ f(t)\,g(t) \xrightarrow{\mathcal{F}} \frac{1}{2\pi}F(\omega)G(\omega) \end{array}}$$

$$f_p(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT_0)$$

$$\Rightarrow F_M(\omega) = \mathcal{F}[f_M(t)] = \frac{1}{2\pi}\mathcal{F}[f(t)] * \mathcal{F}[f_p(t)]$$

$$F_M(\omega) = \frac{1}{2\pi} F(\omega) * F_p(\omega)$$

Donde $F_M(\omega) = F\left[\sum \delta(t - KT_0)\right]$

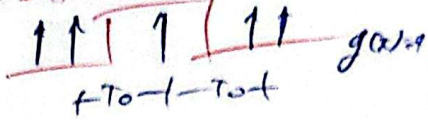
Hallamos:

$$F_p(\omega) = \mathcal{F}\left[\sum_{K=-\infty}^{\infty} \delta(t - KT_0)\right] = 2\pi \sum_{K=-\infty}^{\infty} F_K\, \delta(\omega - K\omega_0)$$

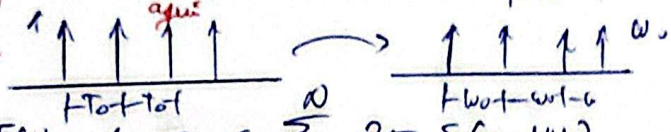
$\omega_0 = \frac{2\pi}{T_0} \Rightarrow$ "Frecuencia de Muestreo"

$\Rightarrow$ Hallamos el coeficiente $\Rightarrow F_K = \frac{1}{T_0}\int_{-\frac{T_0}{2}}^{\frac{T_0}{2}} g(t)\, e^{-jK\omega_0 t}\, dt = \frac{1}{T_0} \Rightarrow$ Porque:

$t \neq 0$ Solo existe aquí



$$\Rightarrow \boxed{F_p(\omega) = \frac{2\pi}{T_0}\sum_{n=-\infty}^{\infty} \delta(\omega - K\omega_0)}$$

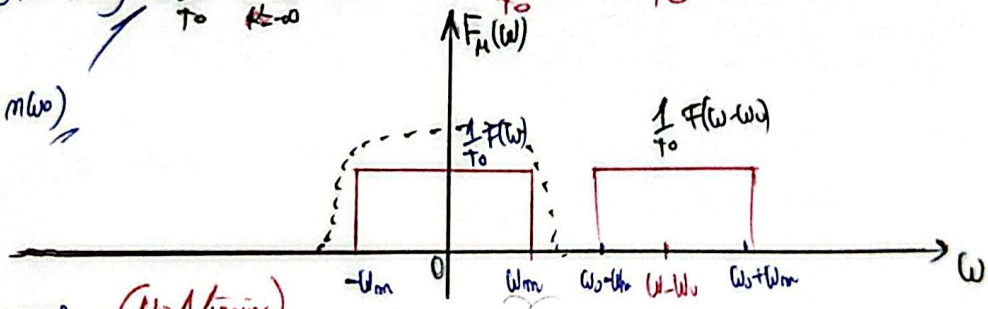


$\Rightarrow$ Regresando: $F_M(\omega) = \frac{1}{2\pi} F(\omega) * F_p(\omega) = \frac{1}{2\pi} F(\omega) * \sum_{n=-\infty}^{\infty} \frac{2\pi}{T_0}\delta(\omega - K\omega_0)$

$$F_M(\omega) = \frac{1}{T_0}\sum_{n=-\infty}^{\infty} F(\omega) * \delta(\omega - K\omega_0) = \frac{1}{T_0}\sum_{K=-\infty}^{\infty} F(\omega - n\omega_0) = \frac{1}{T_0}F(\omega) + \frac{1}{T_0}F(\omega - \omega_0) + \dots$$

Gráfico:

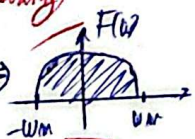
$$F_M(\omega) = \frac{1}{T_0}\sum_{n=-\infty}^{+\infty} F(\omega - n\omega_0)$$



Teorema del Muestreo:

Cantidad mínima de muestras para que la señal se pueda recuperar. → (No Aliasing)

¿Cómo recuperamos la señal Original? →

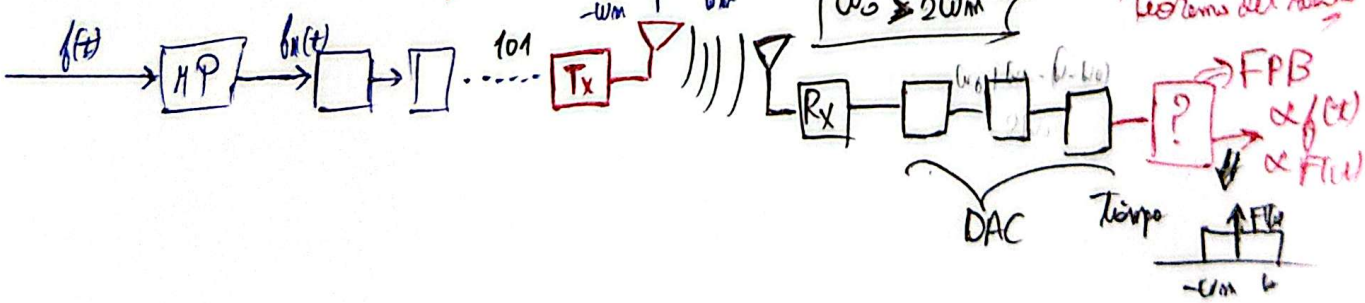


$\omega_m < \omega_0 - \omega_m$

$\omega_0 \geq 2\omega_m$

$\omega_0 \geq 2\omega_m$

Teorema del Muestreo



Escaneado con CamScanner